Fundamental Labs I Week 1 Report

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Week 1 Preparation Assignments

1.

a) Fs is the restorative force needed for the spring to bounce back to its original length after extended. If the spring is extended to the positive direction, x is positive and -kx goes to the negative direction, so as to send it towards its equilibrium position.

b) Fd= -cx appears if there is friction (if there is mass m, connected to a spring of force constant k), is moving on the surface with friction.

If the cart(mass) is moving to the right, the force is to the left, and if the cart is moving to the left, the force is to the right: the direction of the force is the opposite of the direction the cart (mass) is moving into, and the force is velocity dependent. This is why Fd is negative.

2. The motion equation governing the motion of the cart according to Newton's second law

$m\ddot{x}$ = Fs + Fd

If m=1, $m\ddot{x} + c\dot{x} + kx = 0$

$x = e^{\lambda t}$

$$\lambda^2 e^{\lambda t} + c\lambda e^{\lambda t} + k e^{\lambda t} = 0$$

$$e^{\lambda t}(\lambda^2 + c\lambda + k) = 0$$

$$\lambda = \frac{-c \pm \sqrt{c^2 - 4k}}{2}$$

3.

a) λ has a real root if $c^2 - 4k \geq 0$

$$c \geq 2\sqrt{k}$$

(i) If $c^2 - 4k > 0$: Overdamping

The general solution is

$$A = A\exp\left[\left(\frac{-c - \sqrt{4mk}}{2m}\right)t\right]$$

$$+ B\exp\left[\left(\frac{-c + \sqrt{c^2 - 4mk}}{2m}\right)t\right]$$

This equation does not have sine or cosine function (from Euler's formula) , so the motion is not harmonic vibration but overdamped. If the cart (mass) is displaced from x(0)>0 and released from rest, if v(0)=0, then x(t) never becomes 0 and hence cannot become negative. This means the cart(mass) will simply relax to its equilibrium position without any oscillations.

(ii) If $c^2 - 4k = 0$

The motion will be critically damped

General solution :

x= $e^{-\frac{\lambda t}{2}}[A+Bt]$

b) λ has an imaginary root if $c^2 - 4k < 0$

$c < 2\sqrt{k}$

Underdampled motion

General solution:

x= $e^{-\frac{c}{2}t}\left[A\left[\cos\sqrt{k^2-\frac{c^2}{4}}t + i\sin\sqrt{k^2-\frac{c^2}{4}}t\right]\right.$

$\left. + B\left[\cos\sqrt{k^2-\frac{c^2}{4}}t - i\sin\sqrt{k^2-\frac{c^2}{4}}t\right]\right]$

Underdamped and oscillatory motion because of the cos and sine function.

Will oscillate but with a slowly diminishing amplitude.

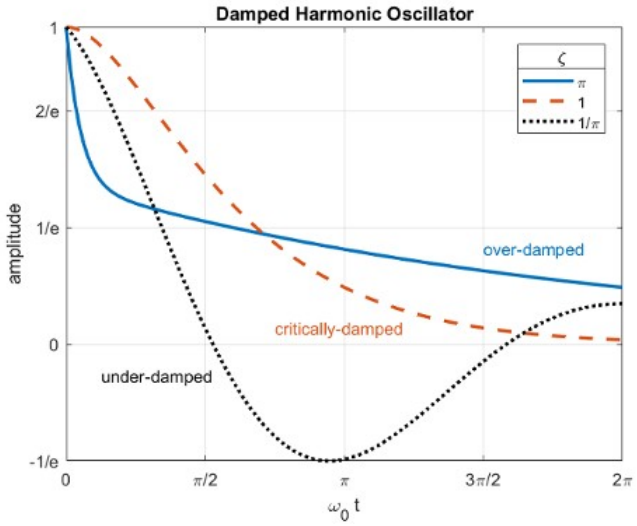


Figure 1 Damped Harmonic Oscillator motions graph from Mathworks

4, 5, and 6 See Figure 1 from MathWorks

Chapter 1 Report Assignment First Week : State equation and feedback control

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Purpose of the lab: To learn the basics of stability and control of dynamical systems through using differential equations, analytical calculation methods, MATLAB and real machine labs

In Chapter 1 in the first week, we learn feedback control on spring-mass-damper system and express state equations for linear systems.

Lab equipment:

1. A mass placed on a horizontal plane such that it moves smoothly in the horizontal direction on a linear bearing, and a viscosity damper equal to an applied spring force in the distance horizontal to that mass. The mass section acts as a DC linear motor (voice 42 coil motor) needle, and can freely apply a thrust that is proportional to the current flowing through the primary side voice coil motor.
2. MATLAB

Reporting assignment 1.1

1. The natural angular frequency $\omega_0 = \sqrt{k/m} = \sqrt{\frac{1.2X10^3}{1.1}}$ Hz = 33.02 Hz

Natural frequency $\omega = \frac{\omega_0}{2\pi}$ = 5.256 Hz = 5.26 Hz

2. If the damping ratio γ is larger or equal to 1, there will be no oscillation.

$$\gamma = \frac{c}{2m\omega_0}$$

If $\gamma \geq 1$, then $c \geq 2m\omega$

c ≥ 2 (1.1) (5.26)

c ≥ 11.572 Nm/s

3. The solution $x = e^{\frac{-ct}{2m}} C \cos\left(\frac{\sqrt{4mk-c^2}}{2m}t + \phi\right)$

If we take the limit as time passes into infinity for the first multiplier $e^{\frac{-ct}{2m}}$,

$\lim_{t\to\infty} e^{\frac{-ct}{2m}} = 0$

So, the displacement of a damper always asymptotically approaches 0 after sufficient time has elapsed from the disposition of the damper and the string.

Reporting assignment 1.2

1. For m =1, the matrix A of the state equation= $\begin{bmatrix} 0 & 1 \\ -k & -c \end{bmatrix}$

   The characteristic equation which contains the eigenvalues can be found from det (A-λI)=0

   Characteristic equation $\lambda^2 + c\lambda + k = 0$

   a) When the characteristic root becomes $\lambda_1 = 2i \; and \; \lambda_2 = -2i$, the characteristic equation becomes
   -4+c(2i)+k=0 and
   -4-c(2i)+k=0
   Solving these simultaneous equations: c=0 and k=4
   In terms of physics from the perspective of the spring force and the damping force, there is no damping force but there is a restorative spring force which is in the direction opposite to the displacement.

   b) If $\lambda_1 = -1 \; and \; \lambda_2 = -1$, the characteristic equation becomes 1+c+k=0 and 1-c-k=0
   Solving these simultaneous equations: c=1 and k=-2
   In terms of physics, there is damping force opposite to the displacement and a spring force that is in the same direction as the displacement. The damping force tries

   c) If λ_1 = 1 + 2i $and \; \lambda_2 = 1 - 2i$, the characteristic equation becomes
   (-3+4i)+(1+2i)+k=0 and (-3+4i)-(1-2i)+k=0
   Solving these simultaneous equations: c=-2 and k=2
   In terms of physics, there is a damping force in the same direction as the displacement and a spring force that is opposite direction to the displacement.

   d) If $\lambda_1 = -1 \; and \; \lambda_2 = -2$, the characterisitic equation becomes
   1-c+k=0 and 4-2c+k=0
   c=3 and k=2
   In terms of physics, the damping force and spring force are both opposite direction to the displacement.

   Sign convention of the spring force and the damping force:
   When the spring is stretched or extended from its free length it will exert a negative force between the connected bodies (between the spring and

the mass) . Likewise when the spring is compressed it will exert a positive force.

When the starting force of a spring is positive it will push apart the two connected bodies. Likewise if the starting force of a spring is negative it will pull together the two connected bodies.

The sign convention of the damping force in the spring is that the damping force always opposes the separation velocity between the two end points of the spring. If the spring is increasing in length then the damping force will push to try to decrease the spring length. Likewise if the spring is decreasing in length then the damping force will push to try to increase the spring length.

Reporting assignment 1.3

1. A= $\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}$

   $\frac{dx}{dt} = A\boldsymbol{x}$ with **x**= $\begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix}$

   $= \begin{bmatrix} \frac{dx}{dt} \\ -\frac{k}{m}x - \frac{c}{m}\frac{dx}{dt} \end{bmatrix}$

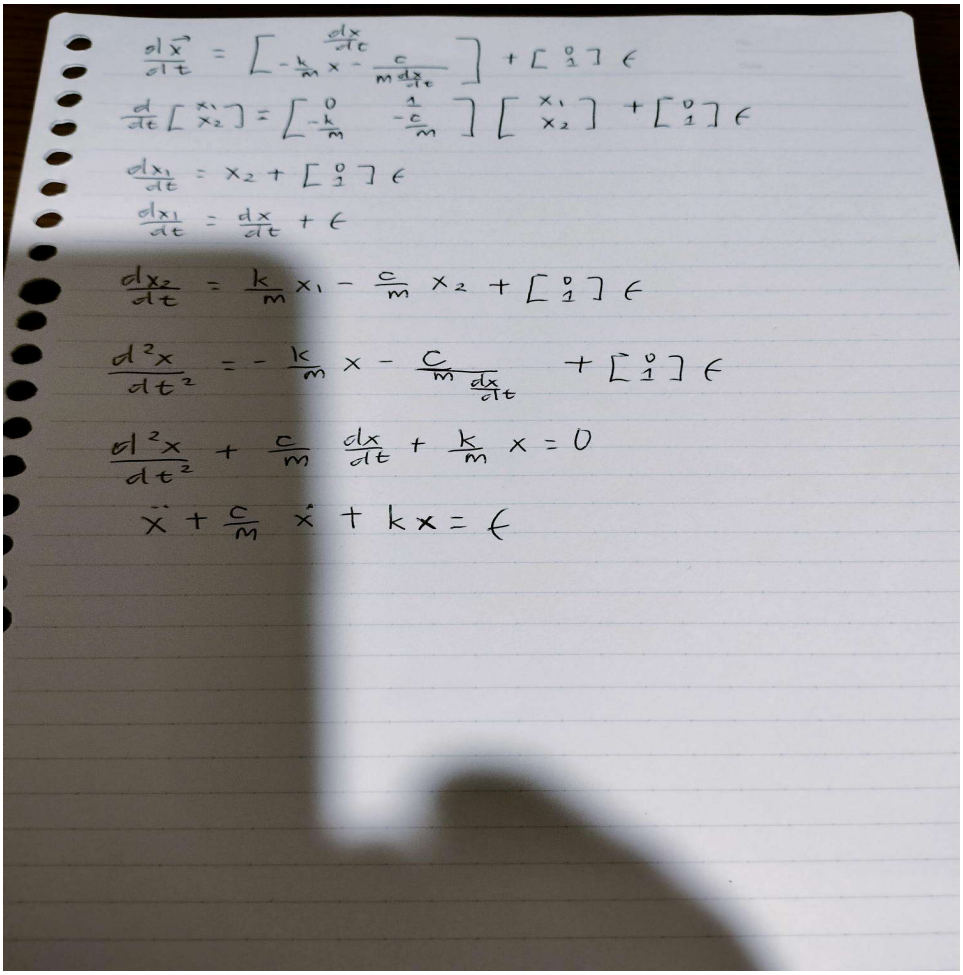
   $$\ddot{x} + (\frac{c}{m})\dot{x} + (\frac{k}{m}) = 0$$
   $$\mathrm{m}\ddot{x} + c\dot{x} + kx = 0$$

   If we use solution x= $e^{\lambda t}$ and substitute it to the former equation, we'll get the secular equation (characteristic equation)
   $m\lambda^2$+cλ+k =0

2. $\frac{dx}{dt} = A\boldsymbol{x} + \boldsymbol{b}u$ with A= $\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}$, **x**= $\begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix}$, **b**= $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and u =f (Equation (1.14)

$$\frac{dx}{dt} = \begin{bmatrix} \frac{dx}{dt} \\ -\frac{k}{m}x - \frac{c}{m\frac{dx}{dt}} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} f$$

By expanding and solving this equation, this matches equation (1.3) $m\ddot{x} + c\dot{x} + kx = f$ (See the figure below)



Hence, the secular equation matches equation (1.14)

Reporting assignment 1.4

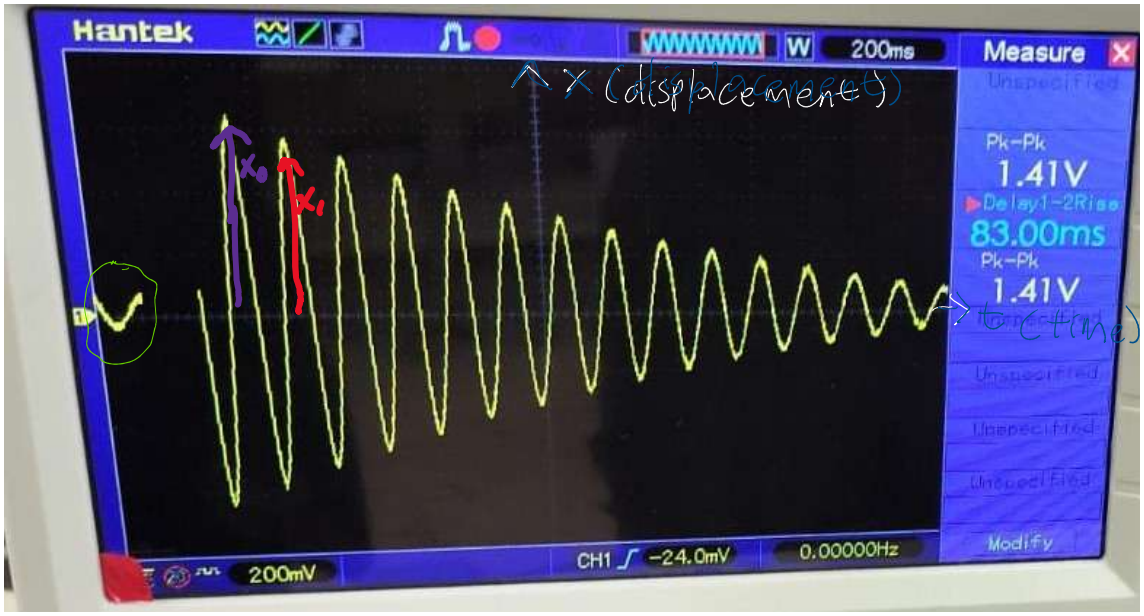


Figure 1 Oscilloscope time-history waveform

The first wave with broken end (marked with green circle) is not included in this wave calculation for ease of calculation

So I will start from this point marked with purple, x_0

1. The system eigenvalues can be seen from the amplitude envelope (which will be the real part of the eigenvalues) and the frequency (which will be the imaginary part of the eigenvalues)

Here, x= p e^{at} $(cosbt + \varphi)$ and $\frac{x_2}{x_1} = e^{a(t_2 - t_1)}$

The system eigenvalues are $\lambda = a \pm ib$

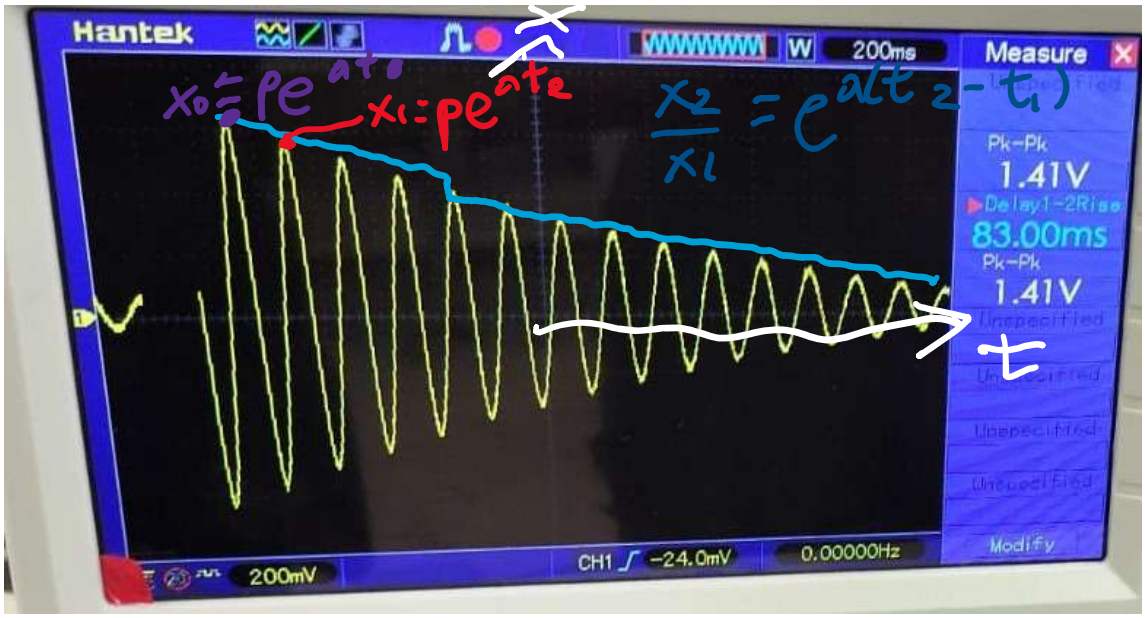


Figure 2

Voltage = 200 mV

From Figure 1, period T = 200 ms

T = $\frac{2\pi}{\omega_0}$, so $\omega_0 = 31.4\ Hz$

2. k= $\omega_0{}^2$ m = (31.4^2) (1.1) = 1085.66 N
3. There is a relationship between logarithmic decrement γ and δ

$$\gamma = \frac{\delta}{\sqrt{\delta^2 + (2\pi)^2}}$$

where $\delta = \ln \frac{x_0}{x_1}$ and x_0, x_1 are amplitudes of two consecutive positive or negative peaks.

From Figure 1, x_0 = (3.2) (10) (2X10^{-3}) = 0.064 mm and $x_1 = (2.8)(10)(2 \times 10^{-3})$

= 0.056mm

$\gamma = \frac{0.134}{\sqrt{(0.134)^2 + (2\pi)^2}} = 0.021$

c = 2m ω_0 $\gamma = 2(1.1)(31.4)$ (0.021)

= 1.45068 =1.45 Ns/m

4. $\lambda = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m} = \frac{-1.45 \pm \sqrt{(1.45)^2 - 4(1.1)(11085)}}{2(1.1)}$

= (-1.45 ± 69.1 i) / 2.2

$\lambda_{1,2} = 0.659 \pm 31.41i$

Reporting assignment 1.5

The state equation of the system is

$$\frac{d}{dt}\begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{-k}{m} - K_p & \frac{-c}{m} - K_d \end{bmatrix} \begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix}$$

So the state equation is $\frac{d}{dt}\begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 986 - K_p & -1.31 - K_d \end{bmatrix}$

2） The block diagram is

